Sahaj Bhadja

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Summary

Software engineer with 2+ years building **C++ systems, Linux applications, networked services, and automated tests**, specializing in concurrency, simulation, performance debugging, client-server architecture, and reliable delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a multi-tenant platform with **normalized schemas, row-level isolation, indexed queries, and validation**, reducing dashboard latency from 1.2 seconds to 350 milliseconds while supporting 500+ daily users.
- Restructured FastAPI workers with **explicit job states, concurrent execution, failure isolation, monitoring, and production debugging**, tripling throughput for 1K+ jobs while preventing cascading failures during peak release cycles.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a concurrent batch-processing engine using **Python multithreading, ThreadPoolExecutor, secure file ingestion, and REST APIs**, processing 1K+ files per batch with fault-tolerant execution across production grading workflows.
- Developed React and Flask client-server features supporting 150+ instructors and 10K+ daily requests, applying **API integration, validation, automated testing, CI/CD, and structured troubleshooting** reliably across peak workloads.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built simulation pipelines processing 100K+ autonomous-driving trajectories, improving planning 18% through **spatial-temporal algorithms, structured data flows, reproducible validation, benchmark analysis, and documented performance analysis** across research experiments.
- Integrated HD-map geometry, lane direction, and ego trajectories with multi-head attention, improving stability 15% while **debugging simulation behavior, analyzing system interfaces, and documenting repeatable engineering test results**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Engineered cross-platform software with **object-oriented modules, microservice interfaces, executable builds, and VM validation**, automating 20+ unit and integration suites consistently across Windows, Linux, and macOS environments reliably.
- Troubleshot large-scale API integrations and ETL workflows using AWS Glue, S3, and SQL, improving response times 30% while **optimizing algorithms, reducing technical debt, and strengthening long-term software maintainability**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **WebSockets, REST APIs, client-server communication, and real-time event handling** across distributed locations.
- Automated Docker and Jenkins build, test, and deployment workflows, reducing release time 50% while applying **Sentry monitoring, structured logging, regression testing, and production issue diagnosis** across services.

Skills

Languages & Systems: C++, C, Python, Java, JavaScript, TypeScript, Linux, Object-Oriented Programming
Networking & Media: TCP/IP, HTTP, Sockets, Client-Server Architecture, RTP, RTSP, GStreamer, WebSockets, REST APIs
Testing: GoogleTest, PyTest, Mockito, Unit Testing, Integration Testing, System Testing, Debugging, Simulation
Build & Delivery: CMake, Git, Jenkins, GitHub Actions, CI/CD, Docker, Shell, Version Control
Engineering: AWS, Azure, GCP, SQL, Microservices, Requirements, SDLC, Documentation, Agile, Scrum

Projects

Embedded Media Transport Node | *C++17, Linux, TCP/IP, GStreamer, POSIX Threads, CMake*

- Designed a **multithreaded embedded media transport service** that accepts RTP streams, buffers audio and video, handles client reconnection, exposes health telemetry, and applies bounded queues for predictable Linux resource usage.

Streaming Systems Test Harness | *C++, Linux, RTP/RTSP, GoogleTest, Wireshark, Python*

- Built a **networked streaming validation harness** generating controlled packet loss, latency, disconnects, and malformed frames, then verifying recovery behavior, interface contracts, resource cleanup, and test results through repeatable scenarios.

CPU-Scheduling Simulator | *C++, React, JavaScript, Operating Systems* | Live

- Implemented **FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue scheduling algorithms** with interactive process controls and Gantt visualization, validating execution correctness and resource-allocation behavior across simulated workloads.